Phase I Out of Distribution Analysis

Individual Contribution Report

Catch Em All Project

Team Vis-AI-Vis
Muhid Syed Muhammad
Student ID k12203959
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Repository Vis-AI-Vis/catch-em-all

Submission Overview

This report presents my Phase I contribution to the Catch Em All project. The work converts the team's preliminary out of distribution analysis into a reproducible local pipeline and examines the rare animal images that the Pokémon classifier predicts with high confidence. The main result is consistent with the team notebook: most foreground animal images receive low confidence, while a small group is mapped confidently to a limited set of Pokémon labels.

Phase I Submission Contents

Based on the team discussion and repository materials, the Phase I package needs the following items:

- A clear research question and short description of the OOD problem.
- Dataset and model details, including the foreground image split and classifier checkpoint.
- A reproducible inference method with confidence, normalized entropy, top two predictions, top two margin, and top five probabilities.
- Overall results for the full foreground set and a separate review of predictions at or above 70 percent confidence.
- Focused visual comparisons for cattle, antelope, and cat images.
- A statement of limitations and a plan for Phase II experiments.
- A record of individual contributions and required disclosure of generative tool use.

My Assigned Work

My assigned work was to support the Phase I analysis with generative tools, document that use, reproduce the classifier analysis locally, compare high and low confidence examples in the three focus categories, and prepare a contribution that the team can review before it is pushed to the shared repository.

Responsibility	Completed output
Repository and environment setup	Local branch with package configuration and documented run commands
Reproducible inference	Batched ViT inference with CSV outputs, plots, and six passing tests
Focused visual review	Cattle, antelope, and cat high versus low confidence montages
Generative tool record	Interaction log, midpoint reflection, and limitations
Submission preparation	This individual contribution report

Research Question

Why do some images within the same real world animal category trigger highly confident Pokémon predictions while most images do not?

Dataset and Model

The analysis uses foreground images from the AM-2k animal matting dataset. The complete team run contains 1,800 images from 20 animal categories. The public folder available for my local reproduction contained 1,766 readable foreground PNG files. Thirty four files listed in the complete split were not present in that download, so local and team totals are reported separately.

The classifier is imzynoxprince/pokemons-image-classifier-gen1-gen9 from Hugging Face. It is a Vision Transformer with a 224 by 224 pixel input and 1,028 output labels in the loaded checkpoint. The classifier is closed-set: every input receives a Pokémon label even when the image is an unrelated animal photograph.

Method

Each foreground image was decoded, processed with the checkpoint's image processor, and passed through the model in batches. The pipeline stores the predicted label, top one confidence, second prediction, top two margin, normalized entropy, top five probabilities, input path, and label count. The 70 percent confidence threshold is an exploratory screening rule, not a validated OOD detector.

Measure	Definition	Use in Phase I
Top one confidence	Largest softmax probability	Finds inputs for which the model strongly favors one label
Normalized entropy	Negative sum of p log p divided by log K	Compares uncertainty across all 1,028 labels on a zero to one scale
Top two margin	Top probability minus second probability	Shows how far the first prediction is ahead of its nearest alternative
High confidence case	Top one confidence at or above 70 percent	Defines the exploratory review subset

Validation

The implementation includes six automated tests covering metric calculations, metadata handling, plot and CSV generation, and model output processing. Image validation found zero decode failures among the 1,766 available foreground files.

Local Reproduction Results

The local run analyzed 1,766 images. Median top one confidence was 10.82 percent and median normalized entropy was 0.770. Only 38 images reached the 70 percent threshold, equal to 2.15 percent of the local set. The distribution therefore has a large low confidence region and a small high confidence tail.

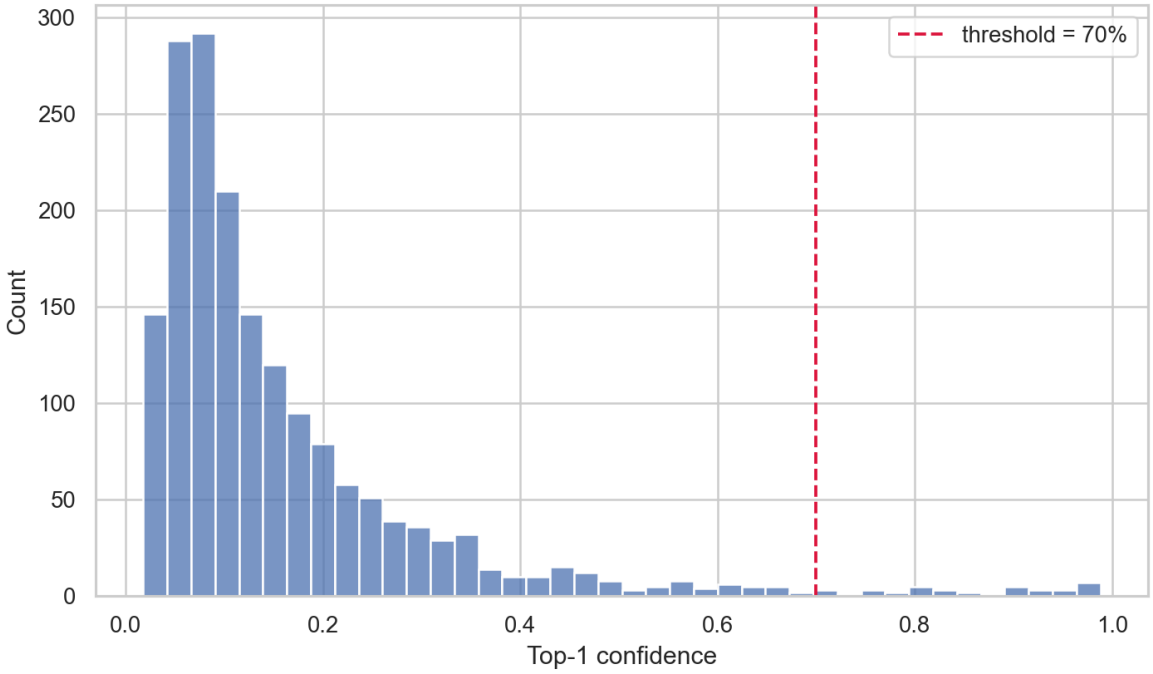


Figure 1. Distribution of top-1 confidence across 1,766 foreground animal images

Interpretation

Most foreground animal images do not produce a dominant Pokémon prediction. The high confidence cases are rare enough to inspect individually, but their scores do not show that the model recognized the input as a Pokémon. They only show that the closed set output distribution favored one available class.

Confidence and Entropy

High confidence cases also tend to have low normalized entropy. This inverse relationship is expected because both measures are calculated from the same probability distribution. The plot is useful for locating the rare bottom right cases, but it is not independent evidence about the cause of misidentification.

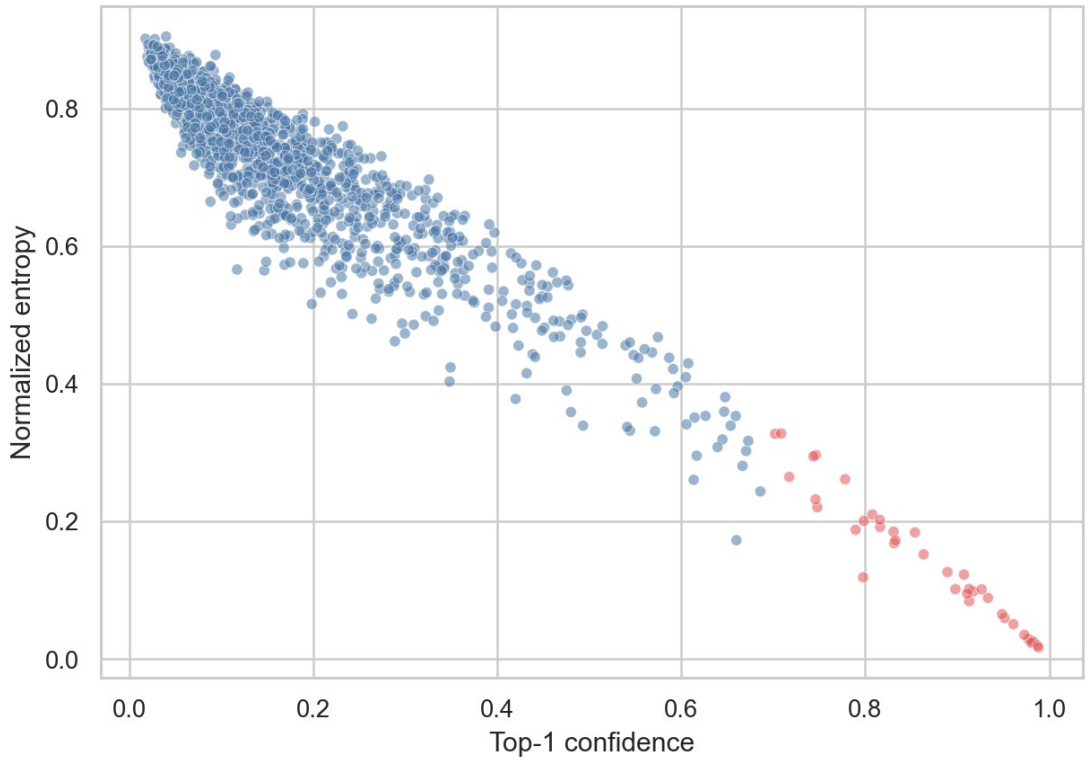


Figure 2. Relationship between top-1 confidence and normalized entropy

High Confidence Labels

Predicted Pokémon	Local count
Bouffalant	17
Sawsbuck	7
Mightyena	6
Linoone	3
Glastrier	2
Mabosstiff	2
Zebstrika	1

Comparison With the Complete Team Run

The complete team notebook reported 39 high confidence images among 1,800 inputs, equal to 2.17 percent. The local reproduction found 38 among 1,766 inputs, equal to 2.15 percent. The close totals and matching dominant labels support the metric and model implementation, while the missing 34 files explain why the two runs should not be treated as identical.

Result	Complete team run	Local reproduction
Images analyzed	1,800	1,766
Images at or above 70 percent	39	38
High confidence rate	2.17 percent	2.15 percent
Output labels	1,028	1,028
Dominant prediction	Bouffalant 17	Bouffalant 17

Complete High Confidence Pairs

Real animal	Predicted Pokémon	Images
Cattle	Bouffalant	17
Antelope	Sawsbuck	7
Cat	Mightyena	4
Dog	Mightyena	2
Leopard	Linoone	2
Antelope	Linoone	1
Antelope	Mabosstiff	1
Deer	Sawsbuck	1
Elephant	Mabosstiff	1
Kangaroo	Glastrier	1
Monkey	Mabosstiff	1
Zebra	Zebstrika	1

Cattle and antelope contribute 26 of the 39 complete high confidence cases. This concentration supports a focused within category comparison rather than a broad claim that all animal images are misclassified in the same way.

Cattle Visual Comparison

The high confidence cattle examples are dominated by shaggy cattle with long lateral or upward curving horns and a centered frontal or three quarter silhouette. Lower confidence examples include calves, short haired cattle, smaller animals in the frame, weak or absent horns, and more varied poses.

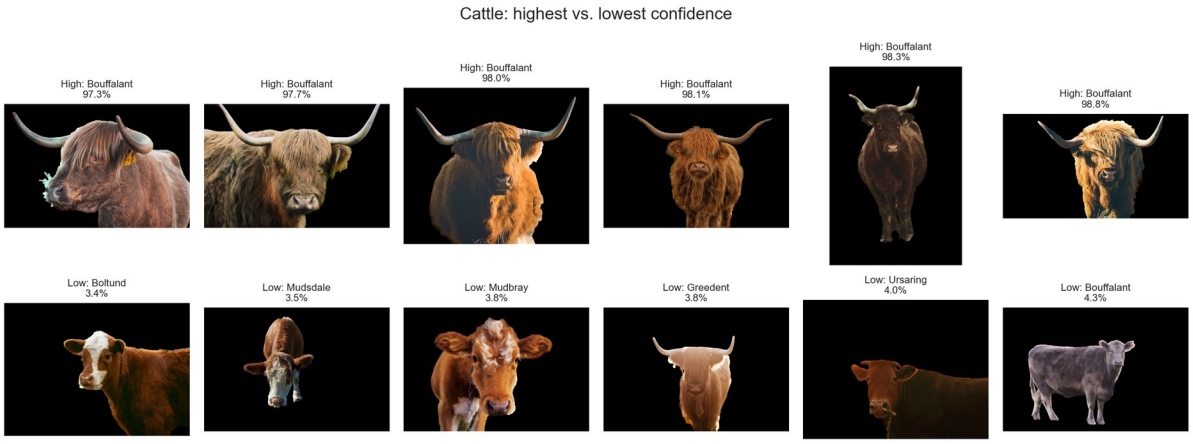


Figure 3. Exploratory high- and low-confidence cattle examples

Pokémon Reference and Interpretation

Pokémon artwork	Model result and interpretation
	**Bouffalant** All 17 high-confidence cattle images in the complete team run were predicted as Bouffalant. The animal and Pokémon share prominent lateral horns, a heavy body, and a dense coat around the head. These similarities make horn geometry and shaggy texture useful Phase II test targets. The visual match is a hypothesis, not proof of which features the model used.

Pokémon artwork: PokéAPI sprites repository. Used for academic comparison.

Antelope Visual Comparison

The high confidence antelope examples usually show a complete tan body, long curved horns, and limited overlap with other animals. Lower confidence examples include close crops, lowered heads, unusual markings, occlusion, or a less complete body outline.

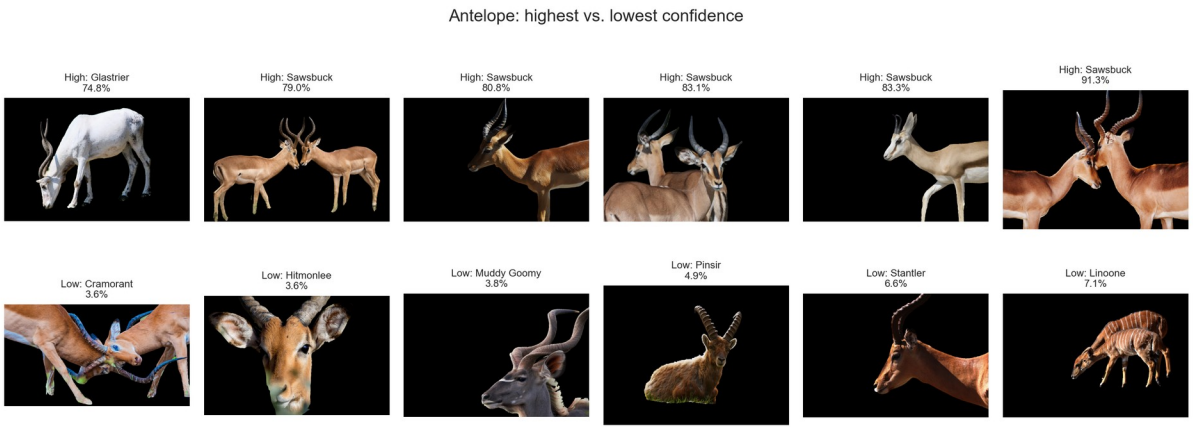


Figure 4. Exploratory high- and low-confidence antelope examples

Pokémon Reference and Interpretation

Pokémon artwork	Model result and interpretation
	**Sawsbuck** Seven of the nine high-confidence antelope images were predicted as Sawsbuck. The remaining two were predicted as Linoone and Mabosstiff. Sawsbuck has a deer-like body and branched antlers, which resemble the full-body antelope examples in the high-confidence group. Body outline and horn geometry are plausible cues, but verified category labels and controlled perturbations are still required.

Pokémon artwork: PokéAPI sprites repository. Used for academic comparison.

Cat Visual Comparison

The high confidence cat examples include strong black and white coat contrast, pointed ears, and a clean isolated silhouette. Lower confidence examples show more varied colors, poses, crops, and foreground extraction artifacts.

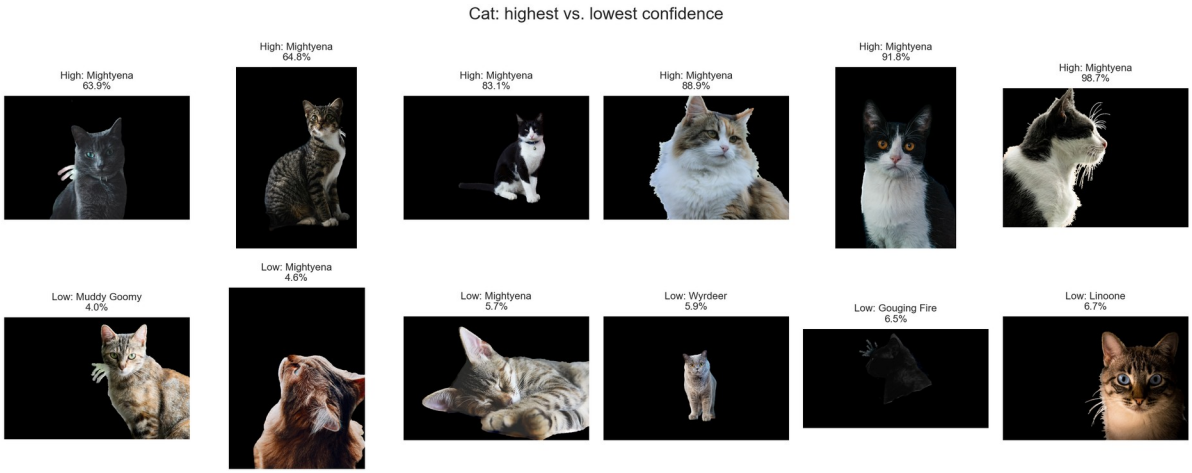


Figure 5. Exploratory high- and low-confidence cat examples

Pokémon Reference and Interpretation

Pokémon artwork	Model result and interpretation
	**Mightyena** All four high-confidence cat images in the complete team run were predicted as Mightyena. The strongest examples show pointed ears, a dark upper coat, pale lower regions, and a clean isolated silhouette. Occluding the ears and reducing coat contrast would test whether those cues affect confidence. The current comparison alone cannot establish causality.

Pokémon artwork: PokéAPI sprites repository. Used for academic comparison.

Other Cat Like Pokémon Checked

Pokémon artwork	Observed lower-confidence alternative
	**Persian** Persian was the top-1 label for 14 of the 90 images in the exploratory local cat group. Its highest confidence was 38.8 percent, and none of the Persian predictions reached the 70 percent threshold. Luxray, Skitty, Delcatty, and Meowth are also visually cat like, but none appeared as a top-1 prediction in this local group. They are therefore noted as visual alternatives rather than observed model outputs.

Pokémon artwork: PokéAPI sprites repository. Used for academic comparison.

Discussion

The Phase I results narrow the problem to a small and structured set of failures. Most foreground animal photographs produce low confidence, while the rare high confidence cases cluster around a few animal and Pokémon pairs. The cattle to Bouffalant and antelope to Sawsbuck patterns are visually plausible, but resemblance alone does not show what the model used internally.

The data support one clear Phase I finding: high-confidence errors cluster around a few animal and Pokémon pairs. Phase II should test which visual cues and ViT layers produce those errors.

Limitations

- The classifier has no explicit OOD rejection class. Confidence describes its closed-set preference, not correctness.
- The 70 percent cutoff is exploratory and has not been calibrated as an OOD threshold.
- The local download contains 34 fewer images than the complete team run.
- Foreground extraction can alter edges, remove context, or leave matting artifacts.
- The public split manifest does not provide category labels. AI-assisted category suggestions are exploratory and imperfect, especially for antelope and sheep.
- Visual comparisons suggest associations but cannot separate horn shape, texture, pose, scale, and extraction quality without controlled experiments.

Phase II Work

- Use verified category metadata for final category rates and within category sampling.
- Apply controlled occlusion and image perturbations to horns, ears, coat texture, and silhouette regions.
- Compare model confidence before and after each perturbation.
- Use ViT compatible attribution methods to identify influential patches and layers.
- Compare the fine-tuned Pokémon model with the base ImageNet model to examine where representations diverge.

Reproducibility and Contribution Record

The work is stored on the local branch syed/phase1-ai-analysis. Commit 3d1b2f1 records the reproducible Phase I implementation. The branch has not been pushed as part of this report. It is ready for team review before remote submission.

File or area	Purpose
src/catch_em_all_phase1/	Inference, metrics, reporting, and visualization package
scripts/pseudo_label_am2k.py	Separate CLIP based suggestions for exploratory category grouping
tests/	Six automated tests for metrics, metadata, and report generation
ai-interactions/	Prompt, output, limitation, and impact log
ai_reflection_midpoint.md	Midpoint reflection on useful and unreliable AI assistance
README.md	Setup, run, tests, and category provenance warning

Reproduction Commands

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e .
catch-em-all-phase1 \
  --images data/am2k/fg \
  --metadata data/am2k/metadata.csv \
  --output outputs/phase1_fg \
  --device auto \
  --batch-size 16
pytest
```

Contribution Status

The local implementation, tests, figures, interaction log, reflection, and submission report are complete. Remaining team action is to review the branch, confirm official category metadata, combine individual work into the group submission, and push the approved changes.

Use of Generative Tools

I used OpenAI Codex to recover task details from the team discussion, inspect the repository, draft parts of the inference pipeline, prepare tests and figures, and organize the documentation. The openai/clip-vit-base-patch32 model produced optional category suggestions for exploratory grouping.

I checked generated outputs against the repository, local test results, model predictions, and the team notebook. I rejected an early row-order category assumption after visual inspection showed mixed categories. I also kept CLIP category suggestions separate from verified labels because the suggestions underdetected antelope and overdetected sheep. The AI-assisted labels are not used as ground truth in the reported team statistics.

Use	Verification	Effect on submission
Recover task and links	Compared with Discord discussion and repository material	Defined my Phase I scope
Draft reproducible code	Ran the pipeline and six automated tests	Produced local predictions and figures
Suggest image groups	Reviewed montages and retained uncertainty fields	Supported exploratory comparisons only
Draft documentation	Checked numbers against CSV outputs and team notebook	Prepared this report and contribution log

Conclusion

My Phase I work reproduces the team's main result and prepares a tested basis for Phase II. In the complete team run, 39 of 1,800 foreground animal images reached 70 percent confidence, with cattle to Bouffalant and antelope to Sawsbuck accounting for most cases. The local run closely matched that pattern on 1,766 available images. The visual comparisons identify testable hypotheses, but causal claims must wait for controlled perturbation and attribution experiments.

References

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